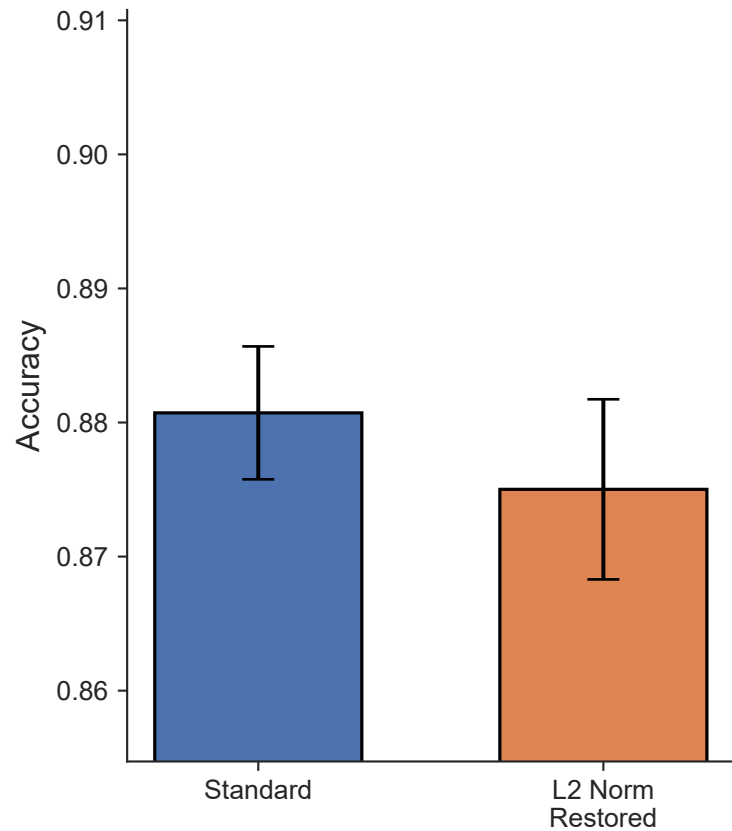


$f_{stage4}$  $f_{refine}$ 